

PolyForge: Joint Cross-Layer Control of Replicas, Semantic Cache, and Model Tier for Multi-Tenant Hybrid Serving

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The problem

- Multi-tenant platforms serve **hybrid** workloads: CRUD + AI inference, same tenants, same cluster.
- Three control surfaces set cost and quality: **replica autoscaling**, **semantic caching**, **model-tier routing**.
- Today: three independent controllers, each locally optimal, none aware of the others — yet they share one budget, one SLO, one tenant set.

Question: do per-layer local optima compose?

Thesis statement

*Per-layer local optima do **not** compose. A controller that plans replicas, cache budget, and model tier **together** — one receding-horizon optimization against $J = \text{cost} + 2 \cdot \text{violation} + 0.5(1 - \text{Jain})$ — dominates tuned per-layer baselines.*

Tested under **7 pre-registered protocols** (pushed before runs), **tuned** baselines, paired seeds, effect sizes with bootstrap CIs — and **published nulls**.

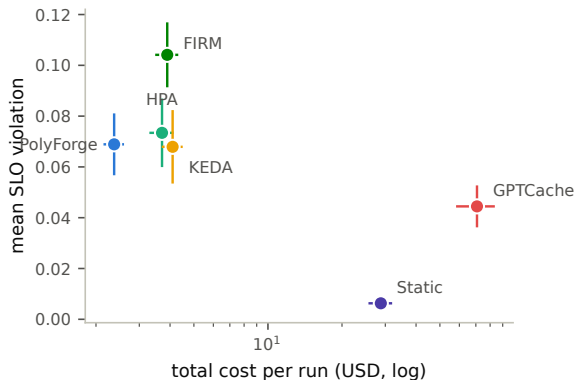
PolyForge in one loop

- **Control plane** (Go): tenants, API keys, RLS-proven isolation, telemetry.
- **AI gateway** (Go): routing, semantic cache with cost-aware eviction.
- **Classifier**: 12-feature windows $\rightarrow$ 5-class taxonomy, drift-checked, every 10 s.
- **JCAC planner**: MPC over a move-blocked lattice (≤ 60 candidates/tenant); the deployed service **imports the simulator's controller** — same code offline and online.
- **Operator**: 4 CRDs; guardrails — replica bounds, budget caps, last-good-plan fallback; shared-target plans are summed, not overwritten.

Methodology: the part built to be attacked

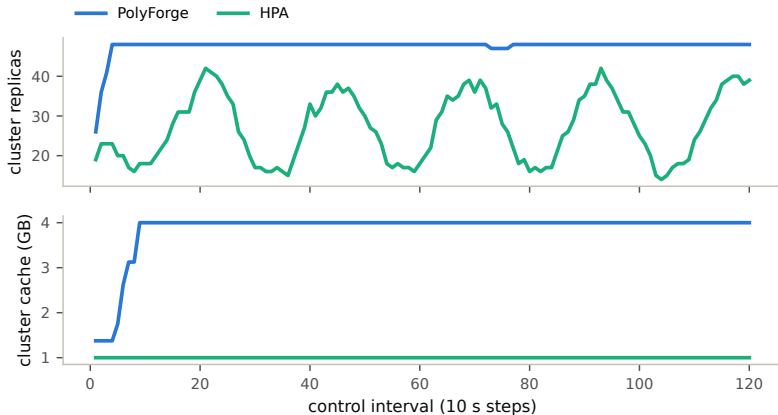
- 7 pre-registrations committed **and pushed before any run**; closed campaigns immutable; stopping rules enforced.
- Baselines (HPA, KEDA, FIRM, GPTCache-LRU, static, VTC-replica, iso-cost) **grid-tuned on our own objective**.
- Paired seeds; citable unit = d_z with 95% bootstrap CI; p-values appear once, as preregates.
- Substrates never mixed; samples never pooled; p95 not p99 (documented).
- **Honest-nulls ledger**: 7 published negatives.

Headline matrix: 1,800 runs, 500 ablations



Wins composite J vs. every tuned baseline ($|d_z|$ 0.66–1.44); only Pareto-undominated system.
Ablations: –jointness = +2884% cost.

Joint control, visible in one trace



First minute of an agentic burst: replicas *and* cache move together; HPA has one knob.

Real demand: BurstGPT replays

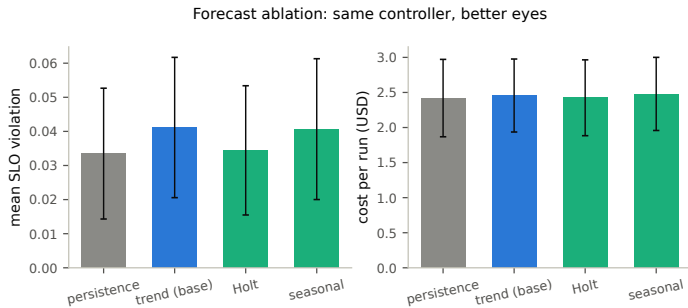
- Real trace: 10.63M requests / 335 days, deterministic ETL.
- First sample $n=16$: **HT FAIL** (underpowered) — point estimates reproduce the v1 ranking. Published as measured.
- Powered replay $n=96$ (new prereg): **HT2 PASS** — J beats HPA/KEDA/FIRM ($p \leq 5.5 \times 10^{-5}$, $d_z -0.43 \dots -0.47$), **cost -70%** ($p \leq 4.3 \times 10^{-6}$), violation trade +0.07 **disclosed as pre-registered**.

Cite -70% ($n=96$), never the underpowered sample's -76%.

The SLO story, told honestly

- v2 H1: beat tuned HPA/KEDA on violation? **FAIL** — parity, at $-43 \dots -48\%$ cost.
- v3 H1' (overload cells): **FAIL** — reactive scalers buy attainment at **3.2–3.4**× **spend**; only PolyForge honors the capacity cap.
- v3 H2': seasonal $<$ trend on violation where periodicity exists: **PASS** ($p = 6 \times 10^{-4}$, no cost premium).
- Spike class (declared exploratory): beats HPA on violation *and* cost ($p = 5 \times 10^{-5}$, -46%).
- Stopping rule: no third confirmatory attempt. The goalposts are in the git history.

Forecasting: the data corrected us



Synthetic stand-in predicted seasonal —44.8% RMSE — **did not transfer** to real BurstGPT (Holt —26.9% wins). Azure LLM 2024 reproduces the boundary: the winner tracks measured periodicity $\Rightarrow$ per-tenant, pluggable forecasting.

Semantic cache on real LMSYS-Chat-1M

- Adaptive hit rate **29.6% @ cosine 0.85** (48.8% @ 0.70); +93% vs. a 10%-fixed cache; \$0.296/1k queries.
- Empirical $h(K)$: $h_{\max}=0.285$, $K_{1/2}\approx 6.6\text{k}$ entries — sim's curve was optimistic; documented, constants untouched.
- **Hit precision measured** (frozen protocol): 0.313 overall @ $\tau=0.85$; same-model $\approx 2\times$ cross-model; near-duplicate ceiling 0.338 $\Rightarrow$ cite the shape, not the absolute.

Fairness and security

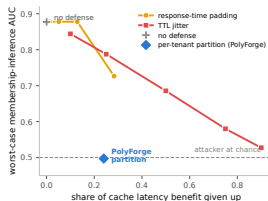
Fairness (control-plane altitude)

- vs. FIRM: Jain 0.9689 / 0.9294 ($d_z=1.0$).
- vs. tuned VTC-replica: wins J ($d_z=-1.12$) and Jain itself (0.9705 / 0.9599) at -35% cost.
- Own γ -term: **published null**.

Security

- Shared cache leaks membership: AUC **0.88** from timing alone.
- Per-tenant partitioning $\rightarrow$ chance (0.50) at 24% latency cost, 76% hits kept — dominates padding & TTL jitter.

Defense frontier: partitioning dominates the known mitigations



Theory, at the code's altitude

- **Prop. 1:** the two-sweep solver returns a block-coordinate minimum of J — no tenant improves unilaterally. (Not claimed: global optimality; Jain coupling is non-separable.)
- **Prop. 2:** self-calibration scale is confined to $[0.5, 1]$, only ever conservative, and returns to full trust in ≤ 25 agreeing steps — safety independent of tuning.
- Measured scale: planner growth exponent **1.45**; 3 s timeout crossed at 128 tenants; past it, designed fallback.

Validity: what bounds the simulator gap

- **Calibration errs against us:** congestion fit $a=0.86$ vs. assumed 1.0; GPU tier bench confirms ordering (1:1.5:16.6).
- **Real data carries the claims:** BurstGPT + Azure demand, LMSYS prompts.
- **Live check is ordinal-only:** HPA arm verified on a real kind cluster; JCAC arm fully wired, gated by an executable actuation check; analysis protocol + figure caption frozen *before* any live number exists.

Not claimed: absolute dollars/latency on production fleets; token-level fairness; p99 on sim.

Contributions

- 1 First joint replicas+cache+tier controller with per-tenant dollar accounting.
- 2 The composition gap, measured: +2884% cost without jointness; -70% cost on real demand vs. tuned scalers.
- 3 Honest SLO account: parity at -43...-48% cost; mechanism confirmed; two failures published.
- 4 Real-data cache results *with a quality denominator*.
- 5 Fairness wins at stated altitude; own term nulled in public.
- 6 Cache side-channel attack + dominant defense frontier.
- 7 Two construction-level solver guarantees.
- 8 A pre-registered, null-publishing systems methodology — in git, diffable.

What remains, and closing

Remaining measurement (one Docker sitting): the live JCAC ordinal slice — protocol, caption, and honesty gate already frozen; reported ordinal-only whichever way it lands.

Future work: token-level demand signals (llm-d/AIBrix substrate), p99 live restatement, planner scaling past 128 tenants, pre-registered adoption of the empirical cache curve.

The strongest sentence in this thesis is not a percentage — it is that every percentage traces to a protocol committed before the run.